

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location STACK SVY02, CA -- station KSJC, America/Los_Angeles
Committed pair OSM 40608200 -> 224174031
Cologix Santa Clara SV1 Data Center / SV Colo
Facade gap 201.2 m between the two halls
Weather record 43,747 real hourly records from KSJC
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.2837 C at 300 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2021-10-22 (chatter)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 7 of 24 hours with 1 mode change. That is 4 more than the reactive incumbent, at 0 unsafe hours against its 2. 6 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 7 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 3 of 24 hours free cooling, 2 mode change(s), 2 unsafe hour(s)
6 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	17.323	18.0	17.220	0.864
01	mechanical	yes	switch budget	17.383	18.0	17.220	0.838
02	mechanical	yes	switch budget	17.877	18.0	18.330	0.678
03	mechanical	yes	switch budget	17.660	18.0	17.780	0.650
04	mechanical	no	dry-bulb	18.154	18.0	18.330	0.805
05	mechanical	no	dry-bulb	18.126	18.0	17.220	0.715
06	mechanical	yes	-	16.850	18.0	14.589	0.664
07	mechanical	yes	-	17.897	18.0	14.581	0.984
08	mechanical	no	dry-bulb	18.457	18.0	14.581	1.204
09	mechanical	no	dry-bulb	18.452	18.0	14.581	1.749
10	mechanical	no	dry-bulb	19.567	18.0	15.701	1.845
11	mechanical	no	dry-bulb	20.222	18.0	17.462	1.559

12	mechanical	no	dry-bulb	20.255	18.0	18.054	1.580
13	mechanical	no	dry-bulb	20.186	18.0	18.336	1.469
14	mechanical	no	dry-bulb	20.533	18.0	18.958	1.410
15	mechanical	no	dry-bulb	19.372	18.0	17.840	1.272
16	mechanical	no	dry-bulb	18.844	18.0	18.390	1.027
17	FREE	yes	-	17.852	18.0	16.682	1.206
18	FREE	yes	-	16.289	18.0	16.110	1.121
19	FREE	yes	-	16.052	18.0	15.001	1.380
20	FREE	yes	-	15.123	18.0	14.508	1.446
21	FREE	yes	-	15.174	18.0	14.441	1.441
22	FREE	yes	-	14.639	18.0	14.440	1.054
23	FREE	yes	-	14.346	18.0	14.446	0.786

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.323 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.383 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.877 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.660 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.154 C against a 18.0 C limit, so it fails by 0.154 C. It would take a limit 0.154 C higher, or a bound 0.154 C tighter, to change this hour.

05:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.126 C against a 18.0 C limit, so it fails by 0.126 C. It would take a limit 0.126 C higher, or a bound 0.126 C tighter, to change this hour.

06:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.850 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the

switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.897 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.457 C against a 18.0 C limit, so it fails by 0.457 C. It would take a limit 0.457 C higher, or a bound 0.457 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.452 C against a 18.0 C limit, so it fails by 0.452 C. It would take a limit 0.452 C higher, or a bound 0.452 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.567 C against a 18.0 C limit, so it fails by 1.567 C. It would take a limit 1.567 C higher, or a bound 1.567 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.222 C against a 18.0 C limit, so it fails by 2.222 C. It would take a limit 2.222 C higher, or a bound 2.222 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.255 C against a 18.0 C limit, so it fails by 2.255 C. It would take a limit 2.255 C higher, or a bound 2.255 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.186 C against a 18.0 C limit, so it fails by 2.186 C. It would take a limit 2.186 C higher, or a bound 2.186 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.533 C against a 18.0 C limit, so it fails by 2.533 C. It would take a limit 2.533 C higher, or a bound 2.533 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.372 C against a 18.0 C limit, so it fails by 1.372 C. It would take a limit 1.372 C higher, or a bound 1.372 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.844 C against a 18.0 C limit, so it fails by 0.844 C. It would take a limit 0.844 C higher, or a bound 0.844 C tighter, to change this hour.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.852 C, which is 0.148 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.206 C, and the margin is measured, not chosen: 0.955 C of group-conditional forecast error for this hour of day, plus 0.2510 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.289 C, which is 1.711 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.121 C, and

the margin is measured, not chosen: 0.943 C of group-conditional forecast error for this hour of day, plus 0.1789 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.052 C, which is 1.948 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.380 C, and the margin is measured, not chosen: 1.163 C of group-conditional forecast error for this hour of day, plus 0.2164 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.123 C, which is 2.877 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.446 C, and the margin is measured, not chosen: 1.323 C of group-conditional forecast error for this hour of day, plus 0.1228 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.174 C, which is 2.826 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.441 C, and the margin is measured, not chosen: 1.262 C of group-conditional forecast error for this hour of day, plus 0.1789 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.639 C, which is 3.361 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.054 C, and the margin is measured, not chosen: 0.860 C of group-conditional forecast error for this hour of day, plus 0.1936 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.346 C, which is 3.654 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.786 C, and the margin is measured, not chosen: 0.608 C of group-conditional forecast error for this hour of day, plus 0.1785 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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